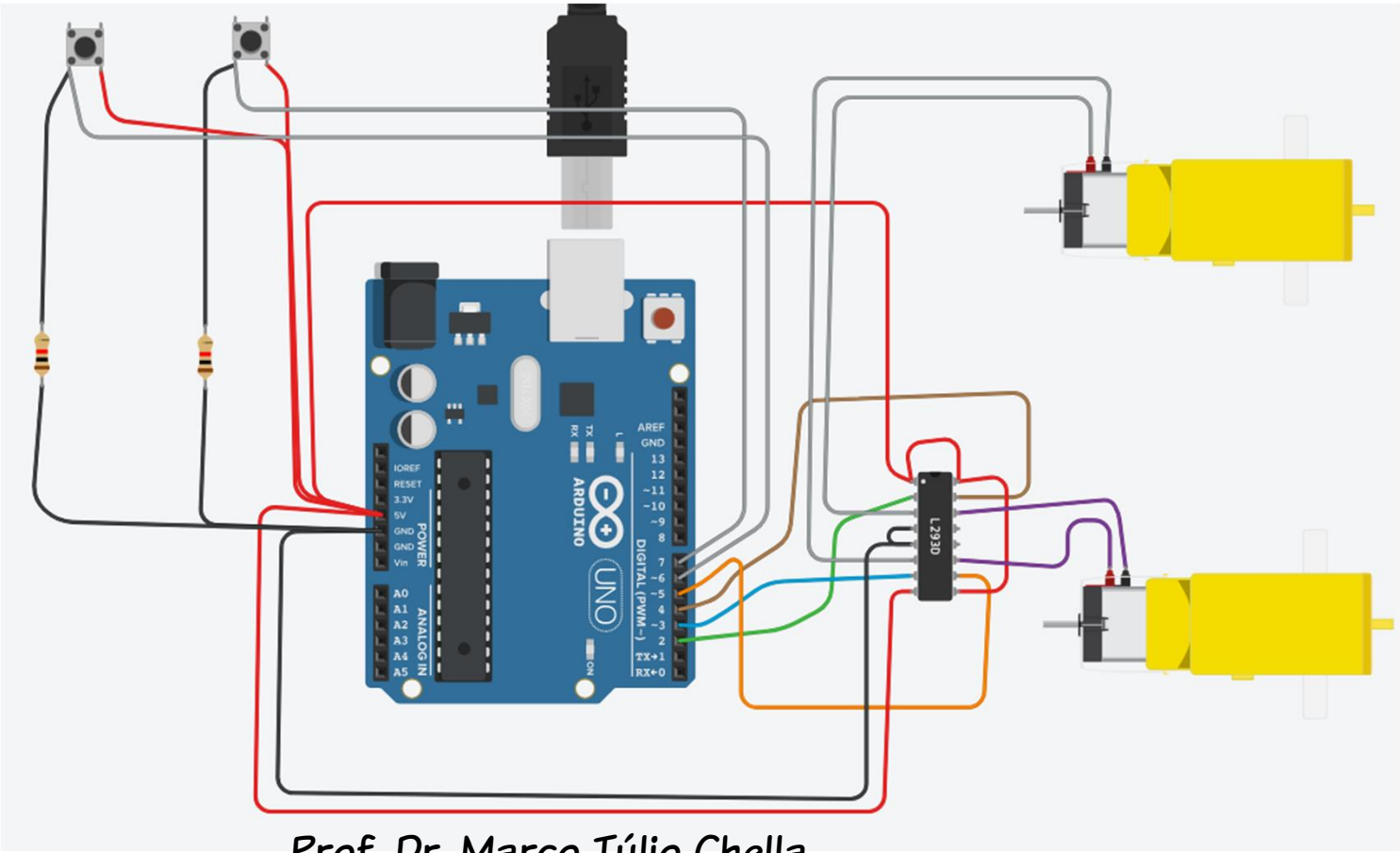
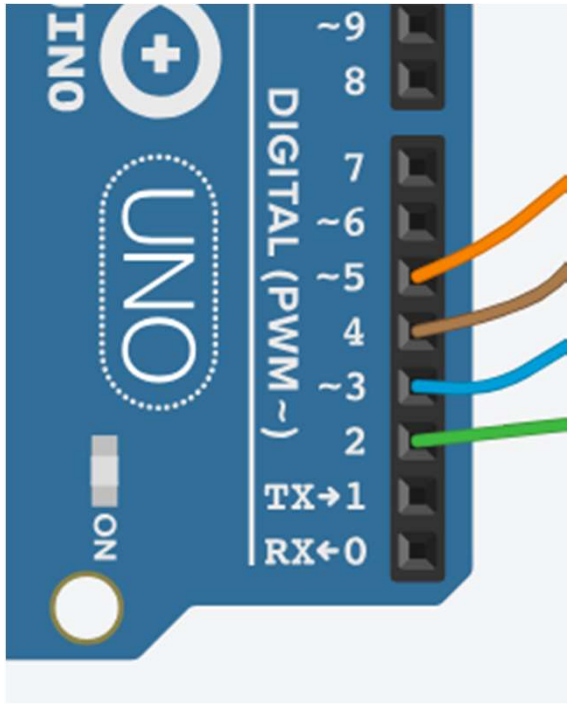
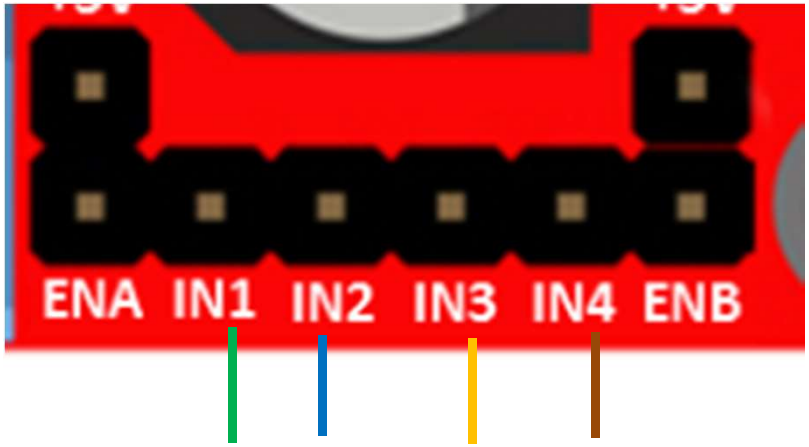
Controle de Motor e Sensor Digital



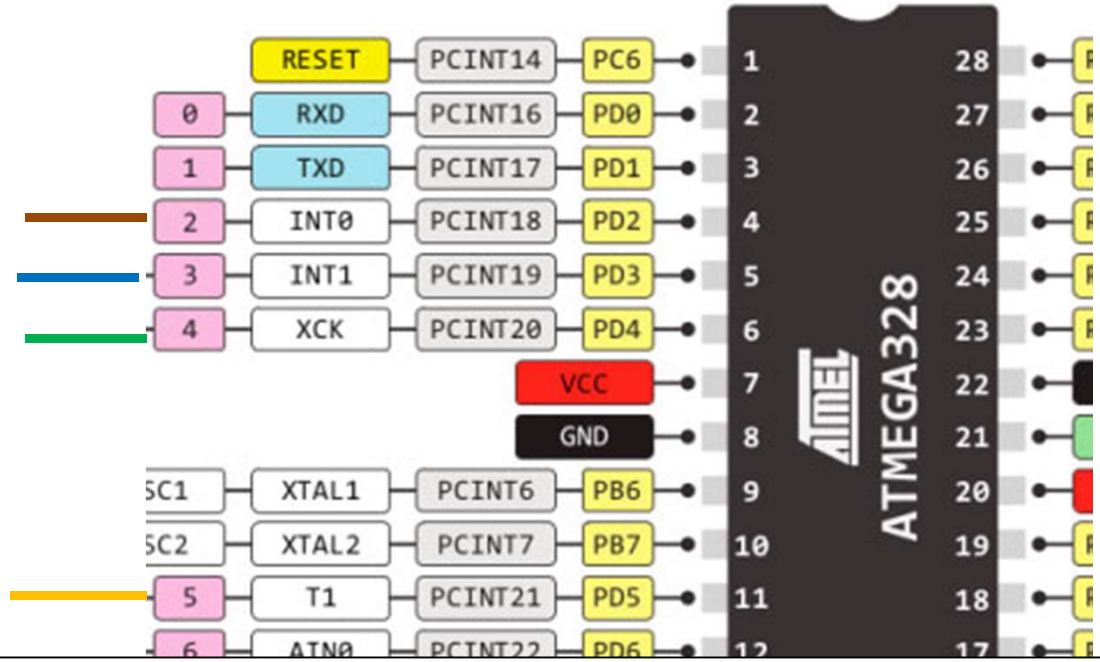
Prof. Dr. Marco Túlio Chella

Motor: Conexões Arduino L298





Motor: Conexões e Configuração



13.4.9 DDRD – The Port D Data Direction Register

Bit	7	6	5	4	3	2	1	0	
0x0A (0x2A)	DDD7	DDD6	DDD5	DDD4	DDD3	DDD2	DDD1	DDD0	DDRD
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Motor: Conexões e Configuração

13.4.9 DDRD – The Port D Data Direction Register

Bit	7	6	5	4	3	2	1	0	
0x0A (0x2A)	DDD7	DDD6	DDD5	DDD4	DDD3	DDD2	DDD1	DDD0	DDRD
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

```
int main (void)
{
 Serial.begin(9600);
DDRD = B00111100;
```

Configuração
Saída digital

Entrada Digital: Conexões e Configuração

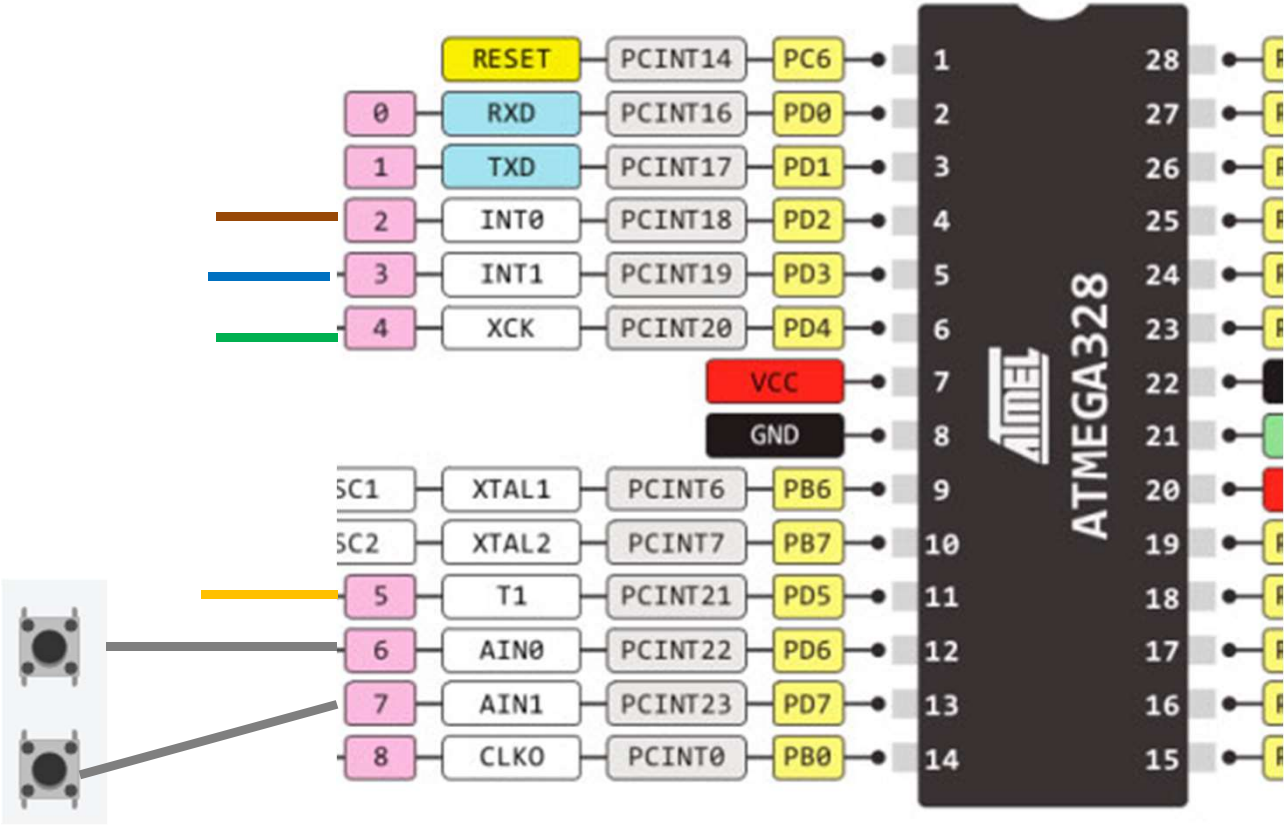
13.4.9 DDRD – The P Da ection Register

Bit	7	6	5	4	3	2	1	0	
0x0A (0x2A)	DDD7	DDD6	DDD5	DDD4	DDD3	DDD2	DDD1	DDD0	DDRD
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

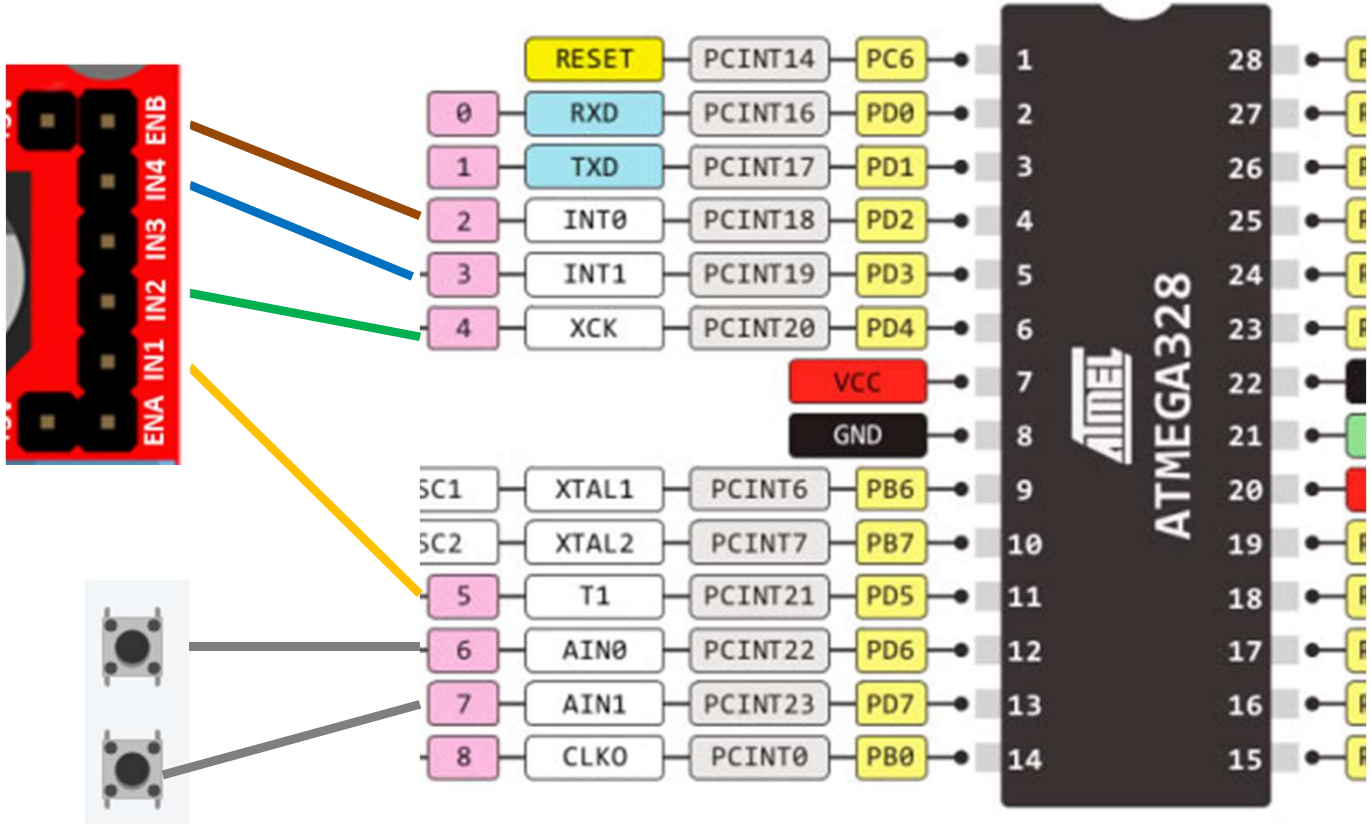
```
int main (void)
{
 Serial.begin(9600);
DDRD = B00111100;
```

Configuração
Entrada digital

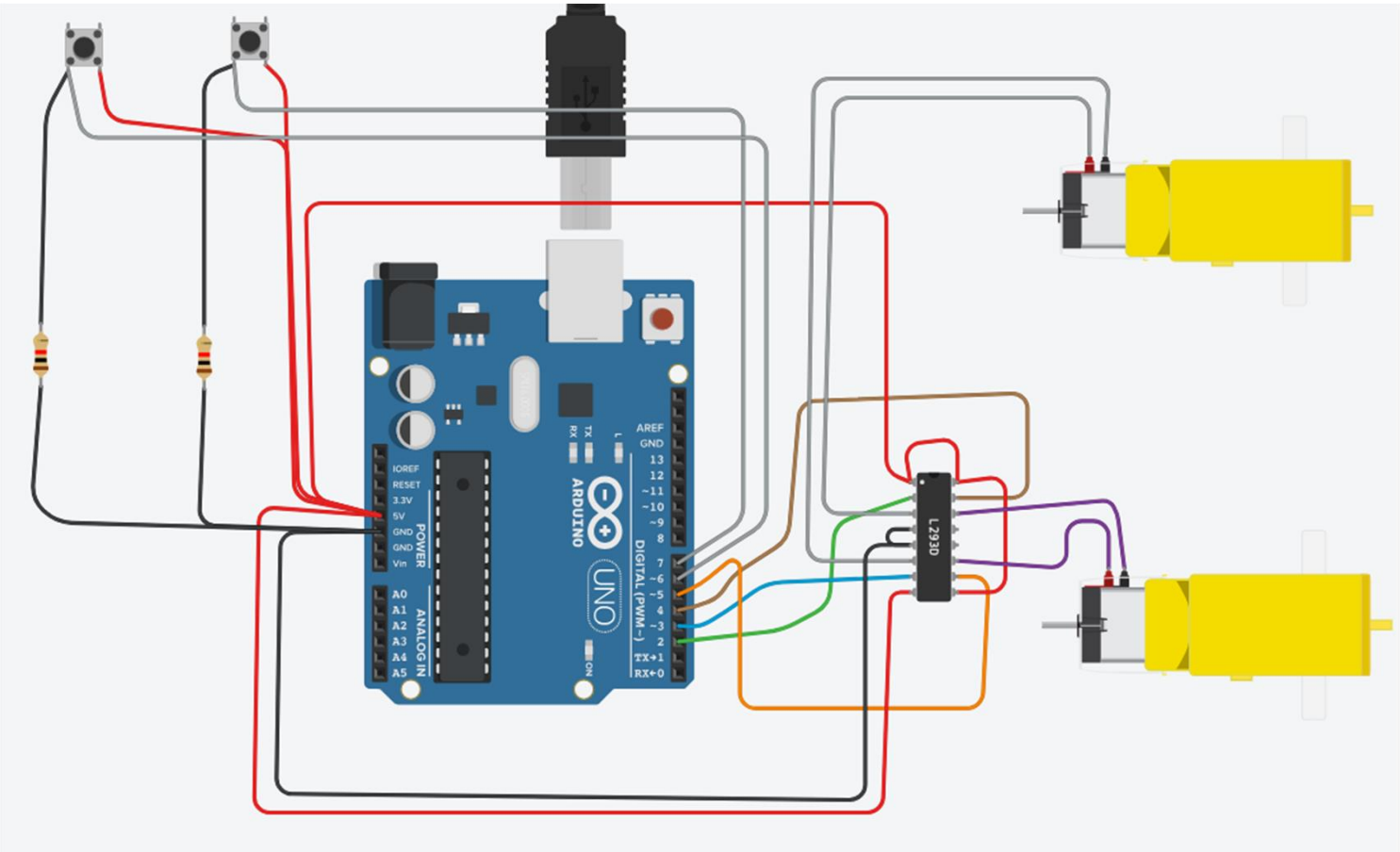
Entrada Digital : Conexões e Configuração



Motor e Entrada Digital : Conexões e Configuração



Entrada Digital e Motor : circuito



Entrada Digital e Motor : Firmware de Teste

```
int main (void)
{
 unsigned char sensor_pino_7;
 unsigned char sensor_pino_6;

 DDRD = DDRD |B00111100;
 Serial.begin (9600);
 while(1) {
 sensor_pino_7 =   (PIND & 128)>> 7;
 sensor_pino_6 =   (PIND & 64)>> 6;
 Serial.println (sensor_pino_6 , DEC);

   if (sensor_pino_7 == 1)
   {
     PORTD = B00101000;
   }

   if (sensor_pino_6 == 1)
   {
     PORTD = B00010100;;
   }

    if (sensor_pino_6 == 0 && sensor_pino_7 == 0 )
    {
      PORTD = B00111100;
    }
}
}
```

Entrada Digital e Motor : Firmware de Teste

```
unsigned char sensor_pino_7;
unsigned char sensor_pino_6;
```

Entrada Digital e Motor : Firmware de Teste

```
sensor_pino_7 =    (PIND & 128)>> 7;
sensor_pino_6 =    (PIND & 64)>> 6;
```

PIND is the input register variable It will read all of the digital input pins at the same time.

Entrada Digital e Motor : Firmware de Teste

Operação E bit a bit

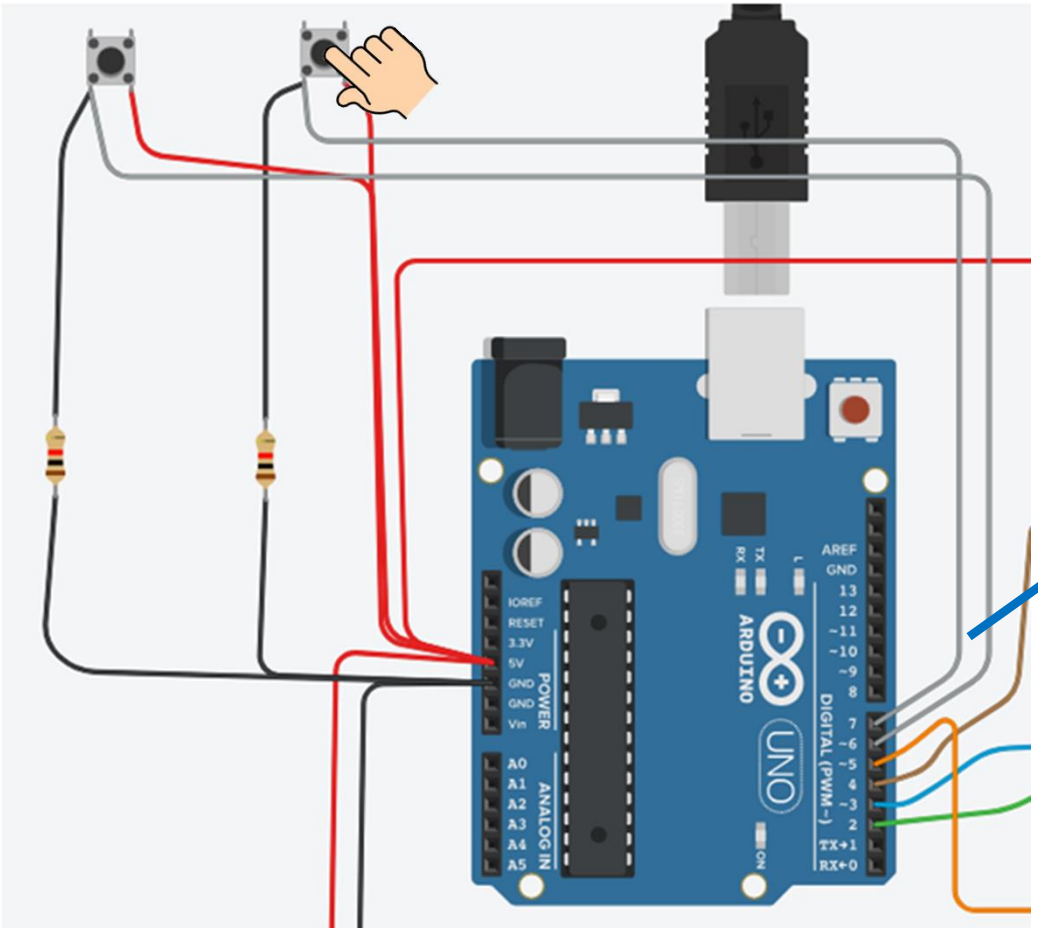
```
sensor_pino_7 =    (PIND & 128)>> 7;
sensor_pino_6 =    (PIND & 64)>> 6;
```

Entrada Digital e Motor : Firmware de Teste

Deslocamento de bit
a direita

```
sensor_pino_7 =    (PIND & 128)>> 7;
sensor_pino_6 =    (PIND & 64)>> 6;
```

Entrada Digital e Motor : Firmware de Teste



5 V => nivel lógico 1

Entrada Digital e Motor : Firmware de Teste

Exemplo:

PIND = 10000000

&10000000 (128 decimal)

10000000

>> 7 00000001

Entrada Digital e Motor : Firmware de Teste

```
sensor_pino_7 =    (PIND & 128)>> 7;
sensor_pino_6 =    (PIND & 64)>> 6;
```

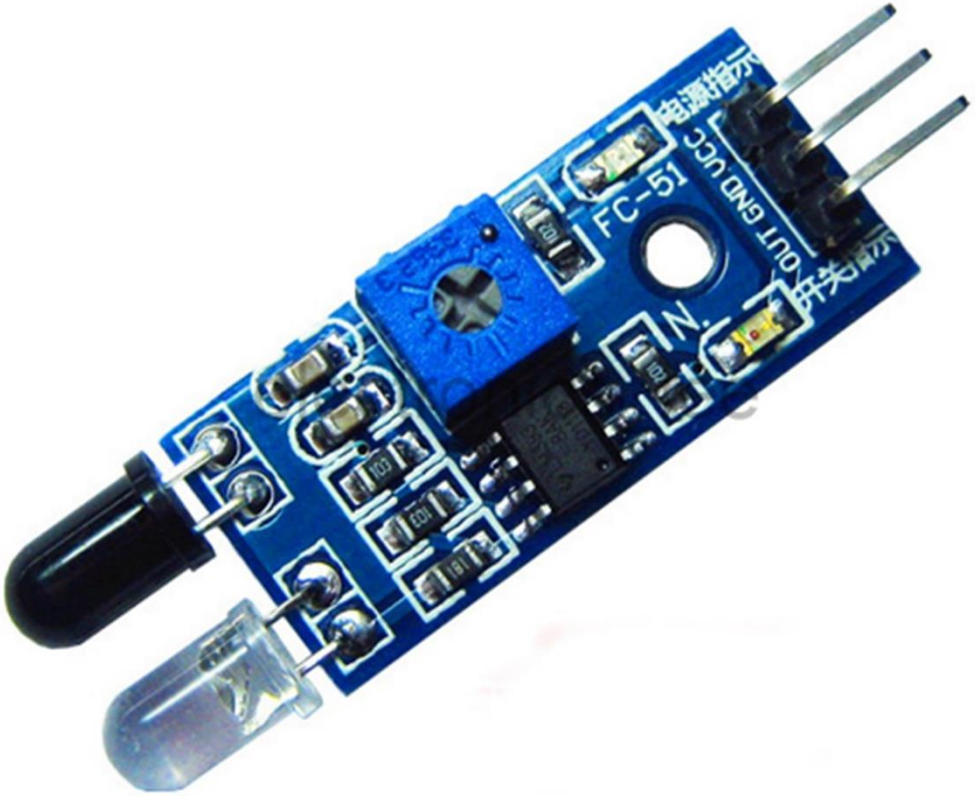
13.4.9 DDRD – The P Da ection Register

Bit	7	6	5	4	3	2	1	0	
0x0A (0x2A)	DDD7	DDD6	DDD5	DDD4	DDD3	DDD2	DDD1	DDD0	DDRD
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

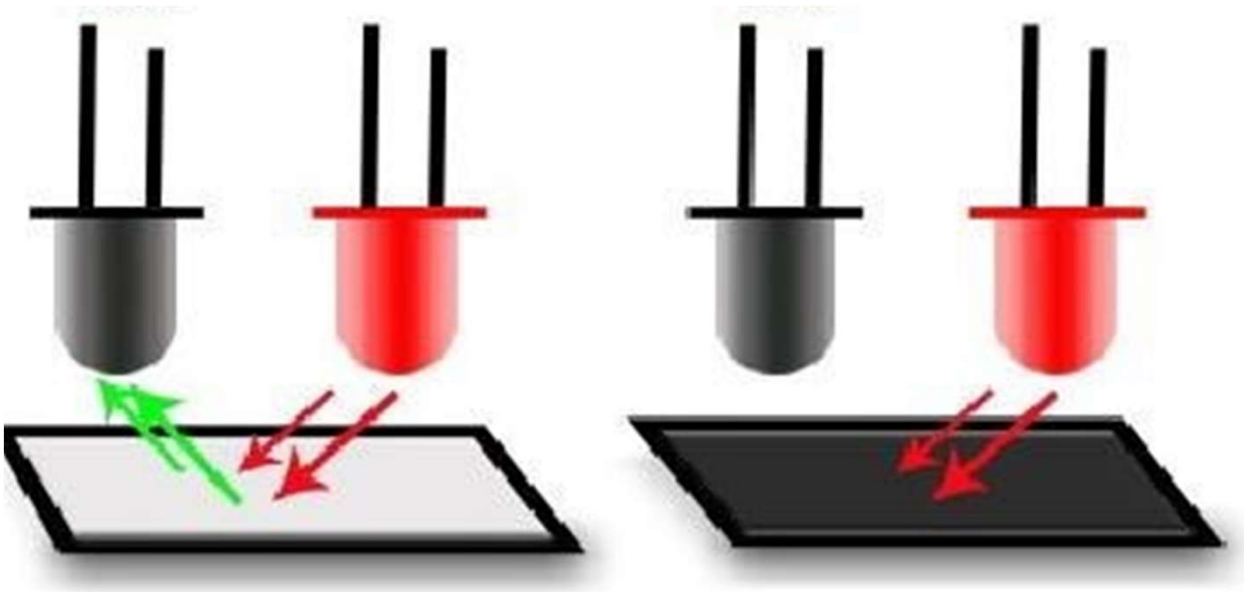
Entrada Digital e Motor : Firmware de Teste

```
if (sensor_pino_7 == 1)
{
  PORTD = B00101000;
}

if (sensor_pino_6 == 1)
{
  PORTD = B00010100;;
}

 if (sensor_pino_6 == 0 && sensor_pino_7 == 0 )
 {
   PORTD = B00111100;
 }
```

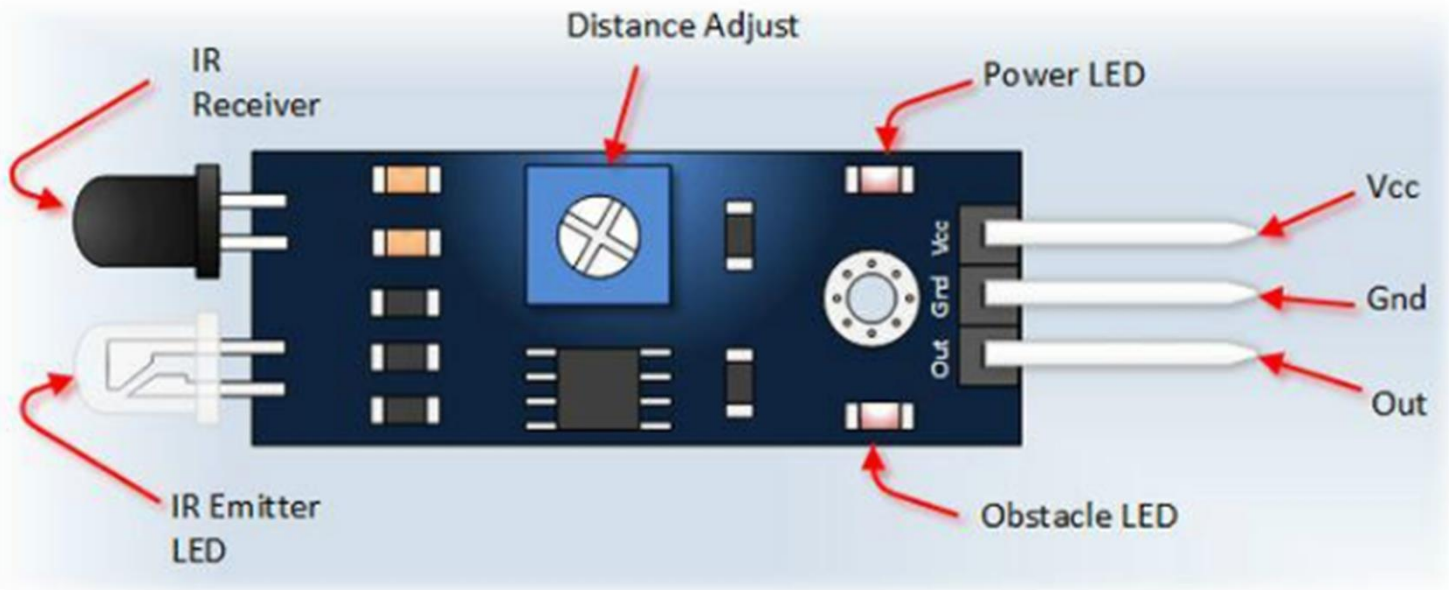
Entrada Digital e Motor : Sensor Infravemelho



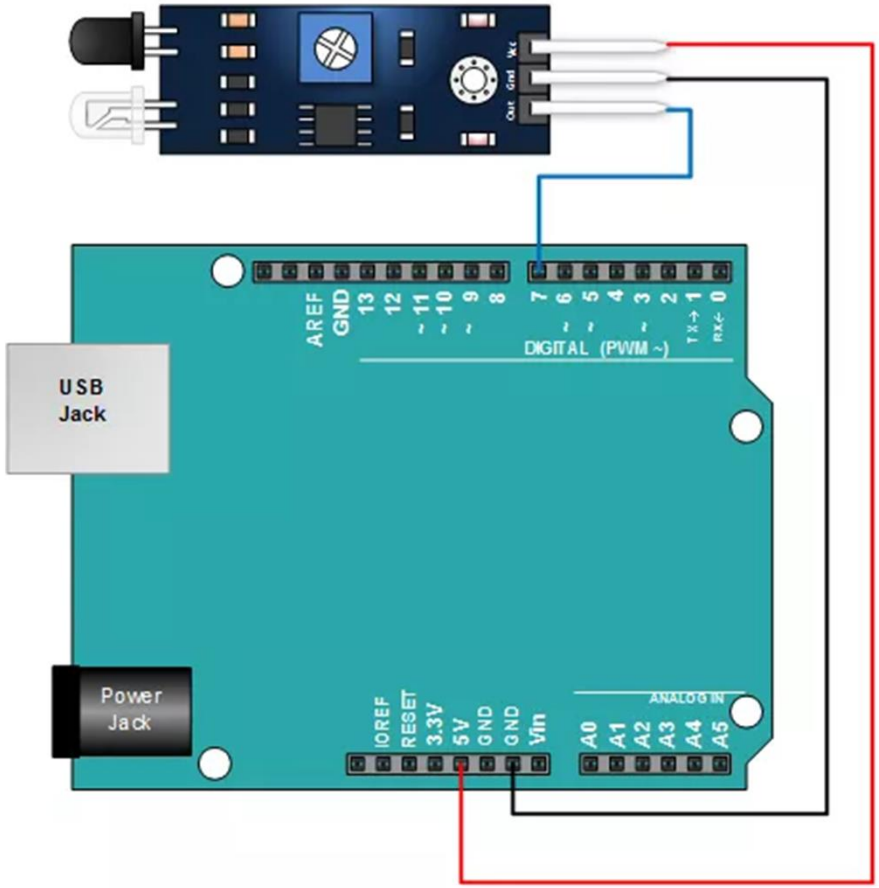
Entrada Digital e Motor : Sensor Infravemelho



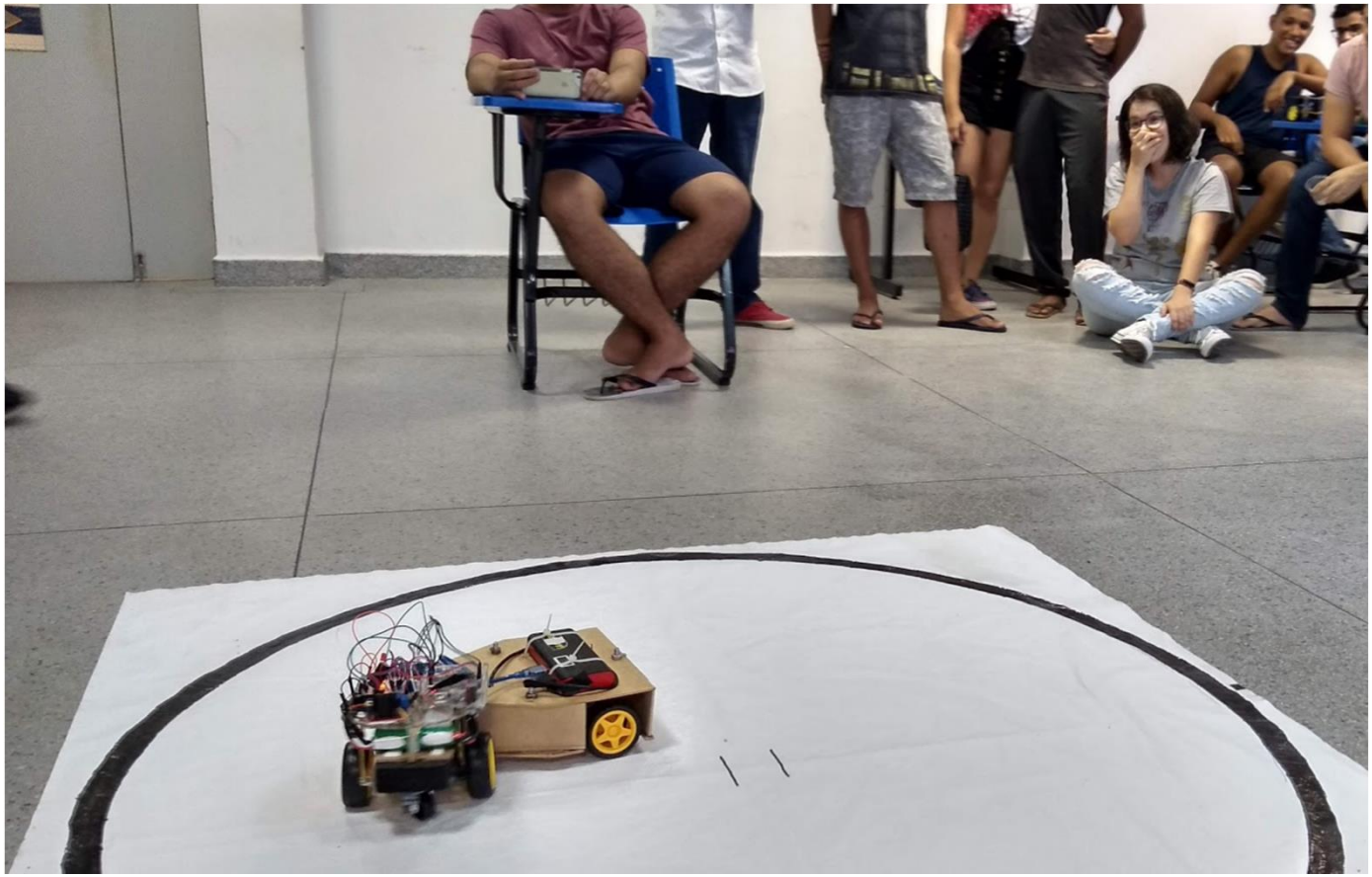
Entrada Digital e Motor : Sensor Infravemelho



Entrada Digital e Motor : Sensor Infravemelho



Entrada Digital e Motor : Sensor Infravemelho



Entrada Digital e Motor : Sensor Infravemelho

Disciplina: Fundamentos de Sistemas Embarcados

2019.2

UFS – Universidade Federal de Sergipe

prof. Dr. Marco Túlio Chella